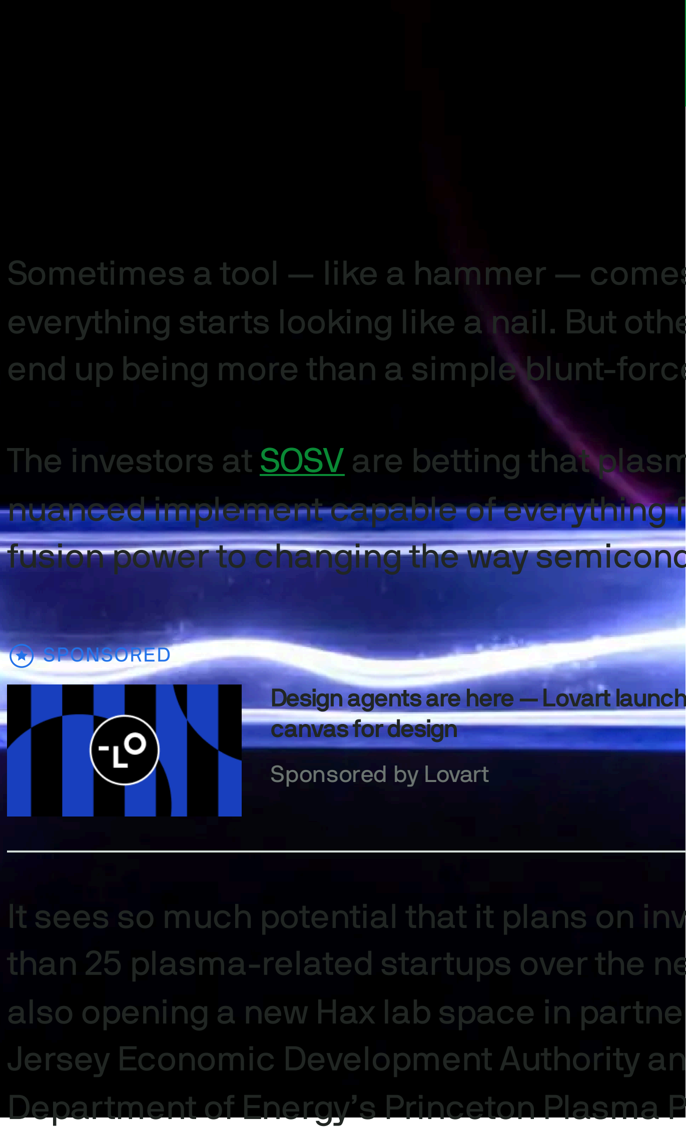
CLIMATE

SOSV bets plasma will change everything from semiconductors to spacecraft

Tim De Chant — 11:26 AM PDT · July 29, 2025



Sometimes a tool — like a hammer — comes along and everything starts looking like a nail. But other times, new tools end up being more than a simple blunt-force object.

The investors at SOSV are betting that plasma will become a nuanced implement capable of everything from enabling fusion power to changing the way semiconductors are made.

It sees so much potential that it plans on investing in more than 25 plasma-related startups over the next five years. It is also opening a new Hax lab space in partnership with the New Jersey Economic Development Authority and the U.S. Department of Energy's Princeton Plasma Physics Laboratory.

IMAGE CREDITS: LYAGOVY / GETTY IMAGES

Nuclear fusion is an obvious place to seed plasma startups. The potential power source works by compressing fuel until it turns into a dense plasma, so dense that atoms begin fusing, releasing energy in the process.

"There's so much here. The best ideas have yet to come to unlock a lot of potential in the fusion space," Duncan Turner, general partner at SOSV, told TechCrunch.

But fusion is just the tip of the iceberg, Turner said. The company has already invested in two companies, including YPlasma, which uses plasma actuators to cool data center chips and control airflow over wind turbine blades.

Finding new uses for plasma in semiconductor manufacturing, where the state of matter already plays a role, could unlock new materials and processes, Turner said. Spacecraft could benefit from plasma thrusters, which are more fuel efficient than traditional thrusters. Harnessing plasma could also unlock entirely new ways to make chemicals like ammonia for fertilizer or fuel from CO_2.

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"All of that combined, we were like, 'Oh wow, there's way more than 25 companies here,'" he said. "There's actually hundreds of companies here."

Topics: Climate sosv

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TRANSPORTATION

IMAGE CREDITS: BRYCE DURBIN/TECHCRUNCH

Tesla shuts down Dojo, the AI training supercomputer that Musk said would be key to full self-driving

Rebecca Bellan — 3:19 PM PDT · August 7, 2025

Tesla is breaking up the team behind its Dojo supercomputer, ending the automaker's play at developing in-house chips for driverless technology, according to Bloomberg.

Dojo's lead, Peter Bannon, is leaving the company, and the remaining team members will be reassigned to other data center and compute projects within Tesla, per Bloomberg's reporting, which cited anonymous sources.

The disbanding of Tesla's Dojo efforts follows the departure of around 20 workers, who left the automaker to start their own AI company called DensityAI. The new startup is reportedly coming out of stealth soon and is building chips, hardware, and software that will power data centers for AI that are used in robotics, by AI agents, and in automotive applications. DensityAI was founded by former Dojo head Ganesh Venkataramanan and ex-Tesla employees Bill Chang and Ben Floering.

It also comes at a crucial time for Tesla.

CEO Elon Musk has pushed to get shareholders to view Tesla as an AI and robotics company, despite a limited robotaxi launch in Austin this past June that featured Model Y vehicles with a human in the front passenger seat and resulted in a number of reported incidents of the vehicles exhibiting problematic driving behavior.

Tesla's decision to shut down Dojo, which Musk has been talking about since 2019, is a major shift in strategy. Musk has said that Dojo would be the cornerstone of Tesla's AI ambitions and its goal to reach full self-driving due to its ability to "process truly vast amounts of video data." He talked about Dojo, albeit briefly, as recently as the company's second-quarter earnings call.

In 2023, Morgan Stanley predicted Dojo could add $500 billion to the company's market value by unlocking new revenue streams in the form of robotaxis and software services. Just last year, Musk noted that Tesla's AI team would "double down" on Dojo in the lead-up to Tesla's robotaxi reveal, which happened in October.

But talk about Dojo halted around August 2024, when Musk began touting Cortex instead, Tesla's "giant new AI training supercluster being built at Tesla HQ in Austin to solve real-world AI."

The Dojo project was one part supercomputer, one part in-house chip-making. Tesla unveiled its D1 chip when it formally announced Dojo at its first AI Day in 2021. Venkataramanan presented the chip, which Tesla said would be used alongside Nvidia's GPU to power the Dojo supercomputer. The automaker also said it was working on a next-gen D2 chip that would solve any information flow bottlenecks of its predecessor.

Sources told Bloomberg that now Tesla plans to increase its reliance on Nvidia, as well as other external tech partners like AMD for compute and Samsung for chip manufacturing. Tesla last month signed a $16.5 billion deal with Samsung to make its AI6 inference chips, a chip design that promises to scale from powering FSD and Tesla's Optimus humanoid robots all the way to high-performance AI training in data centers.

During Tesla's second-quarter earnings call, Musk hinted at potential redundancies.

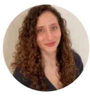
"Thinking about Dojo 3 and the AI6 inference chip, it seems like intuitively, we want to try to find convergence there, where it's basically the same chip," Musk said.

The news comes as Tesla's board offers Musk a $29 billion pay package to keep him at Tesla and help push the company's AI efforts forward, rather than getting too sidetracked by his other companies, including the more pure-play AI startup xAI.

TechCrunch has reached out to Tesla for more information.

Have a sensitive tip or confidential documents? We're reporting on the inner workings of the AI industry — from the companies shaping its future to the people impacted by their decisions. Reach out to Rebecca Bellan at rebecca.bellan@techcrunch.com and Maxwell Zeff at maxwell.zeff@techcrunch.com. For secure communication, you can contact us via Signal at @rebeccabellan.491 and @mzeff.88.

Topics: AI autonomous vehicles Dojo Elon Musk robotaxis Tesla Transportation



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SPACE

Rocket Lab eyes big defense opportunities with new acquisition

Rocket Lab's second-quarter results, which were posted Thursday, show revenues continue to be driven by its space systems business rather than launch. The results also highlighted the company's acquisition strategy and how its purchase of a new optical payloads company will make it more competitive for lucrative government contracts.

The company's space systems brought in $97.9 million of the $144.5 million in total revenues for the second quarter. Rocket Labs' total revenue, its highest quarterly revenue in the company's history, jumped 36% from a year ago. The company's net loss widened to $66.4 million.

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Rocket Lab founder and CEO Peter Beck highlighted the "busy quarter of M&A activity" as the company gets close to closing its deal to buy Geost, a company that builds optical payloads. Through that acquisition, which will close at $275 million in cash and equity, Rocket Lab is opening a new business unit called Optical Systems to scale electro-optical and infrared sensor manufacturing.

Those sensors are used in missile warning, tracking, and space domain awareness. The acquisition is also one part of a larger play from Rocket Lab — spelled out quite explicitly in the company's earnings presentation — to bid for multibillion-dollar DOD initiatives like Golden Dome.

Rocket Lab has already won some major defense contracts, including a $515 million, 18-satellite build contract to support the Space Development Agency's constellation of missile-tracking satellites. The company has moved into production on those satellites, following confirmation that they meet the DOD's mission requirements, Rocket Lab said.

Down on Earth, Rocket Lab says it's progressing well toward the first launch of its larger Neutron rocket. The rocket's launch complex in Virginia is expected to be complete in the third quarter of this year, with hardware en route and the new rocket engine, called Archimedes, undergoing multiple tests per day.

The company is staying vague about the exact launch date, saying only in the presentation that it is pushing an "all-out

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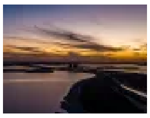

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effort to get Neutron to the launch pad before the end of 2025."

Rocket Lab ended the quarter with $564 million in cash and cash equivalents and estimates revenues will hit between $145 to $155 million next quarter.

Topics: earnings Mergers and Acquisitions Rocket Lab Space

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STARTUPS

High costs and thin margins threatening AI coding startups

Marina Temkin · 2:05 PM PDT · August 7, 2025

In February, AI coding startup Windsurf was in talks to raise a big new round at a $2.85 billion valuation led by Kleiner Perkins, at double the valuation it hit six months earlier, sources told TechCrunch at the time. That deal didn't happen, according to a source familiar with the matter. Instead, news broke in April that the startup planned to sell itself to OpenAI for roughly the same valuation: $3 billion.

IMAGE CREDITS: YUICHIRO CHINO / GETTY IMAGES

While that deal famously fell apart, one bigger question remains: If the startup was growing that fast and attracting VC interest, why would it sell at all?

Insiders tell TechCrunch that for all the popularity and hype around AI coding assistants, they can actually be massively money-losing businesses. Vibe coders generally, and Windsurf in particular, can have such expensive structures that their gross margins are "very negative," one person close to Windsurf told TechCrunch. Meaning it cost more to run the product than the startup could charge for it.

This is due to the high costs of using large language models (LLMs), the person explained. AI coding assistants are particularly pressured to always offer the most recent, most advanced, and most expensive LLMs because model makers are particularly fine-tuning their latest models for improvements in coding and related tasks like debugging.

This is a challenge compounded by fierce competition in the vibe-coding and code-assist market. Rivals include companies that already have huge customer bases like Anysphere's Cursor and GitHub Copilot.

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The most straightforward path to improving margins in this business involves the startups building their own models, thereby eliminating costs of paying suppliers like Anthropic and OpenAI.

"It's a very expensive business to run if you're not going to be in the model game," said the person.

But that idea comes with its own risks. Windsurf's co-founder and CEO, Varun Mohan, ultimately decided against the company building its own model — an expensive undertaking, the person said.

In addition, model makers are already competing directly. Anthropic offers Claude Code and OpenAI offers Codex, for instance.

Selling the business was a strategic move to lock in a high return before it could be undermined by the very companies that supplied its AI, including OpenAI and Anthropic, which were also entering the AI coding market.

Multiple people believe that the same pressure on margins Windsurf faced could be impacting Anysphere, the maker of Cursor, as well as vibe coders like Lovable, Replit, and others.

"Margins on all of the 'code gen' products are either neutral or negative. They're absolutely abysmal," said Nicholas Charriere, founder of Mocha, a vibe-coding startup and back-end hosting solution serving small and medium businesses

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(SMBs). He added that he believes the variable costs for all the startups in the sector are very close, likely within 10% to 15% of one another.

Unlike Windsurf, Anysphere has been growing so fast that it intends to remain an independent company, having already turned down acquisition offers, including, reports say, from OpenAI.

And Anysphere announced in January that it is attempting to build its own model, which could give it more control over its expenses. In July, the startup hired two leaders from Anthropic's Claude Code team, the Information reported, but two weeks later, these employees returned to work at Anthropic.

In addition to building a model, Anysphere could expect the cost of LLMs to decrease over time.

"That's what everyone's banking on," said Erik Nordlander, a general partner at Google Ventures. "The inference cost today, that's the most expensive it's ever going to be."

It's not entirely clear how true that is. Rather than falling as expected, the cost of some of the latest AI models has risen, as they use more time and computational resources to handle complicated, multistep tasks.

When that will change remains to be seen. On Thursday, for instance, OpenAI introduced a new flagship model, GPT-5, with fees that are significantly less than its competitor, Anthropic's Claude Opus 4.1. And Anysphere immediately offered this model as a choice for Cursor users.

Anysphere has also recently changed its pricing structure to pass along the increased costs of running Anthropic's latest Claude model, particularly to its most active users. The move caught some of Cursor customers by surprise, since they didn't expect additional charges on top of its $20-per-month Pro plan. Anysphere CEO Michael Truell later apologized for unclear communication about the pricing change in a blog post.

This is the rock and the hard place. Although Cursor is one of the most popular AI applications, having reached $500 million in ARR in June, the company's user base may not be so loyal to the product if another company develops a tool that is superior to Cursor, investors say.

Anysphere didn't respond to a request for comment.

Given the competitive landscape and the costs, Windsurf's decision to get out may prove to be understandable. After the OpenAI deal fell through, the founders and key employees left to join Google in a deal that led to a $2.4 billion payout to key shareholders. The remaining business then sold itself to Cognition.

While many, including prominent VCs, criticized Mohan for leaving approximately 200 employees without roles at Google, a source familiar with the deal insisted the acquisition actually maximized the outcomes for all employees.

Beyond Cursor, other AI coding tools are also among the fastest growing startups of the LLM generation, like Replit, Lovable, and Bolt, and all of them rely on model makers as well.

Additionally, if this extremely popular business sector, already generating hundreds of millions in revenue or more a year, has difficulty building on top of model makers, what might it mean for other, more nascent industries?

Topics: AI coding assistants cursor Startups Venture Windsurf

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